

SEMESTER END EXAMINATIONS – MARCH 2024

Program	: B.E. – Computer Science and Engineering	Semester	: III
Course Name	: Data Structures	Max. Marks	: 100
Course Code	: CS33	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT – I

- With the help of program, bring out the difference between malloc, calloc and realloc dynamic memory functions. CO1 (09)
 - What is Abstract Data Type? Write an ADT for strings. CO1 (06)
 - Write a simple transpose function of a Sparse Matrix. CO1 (05)
- With the help of program explain the concept function like macros. CO1 (05)
 - What is an algorithm? List the five criteria to design a good algorithm. CO1 (06)
 - Write a C program to perform addition of two polynomials using an array of structures. CO2 (09)

UNIT – II

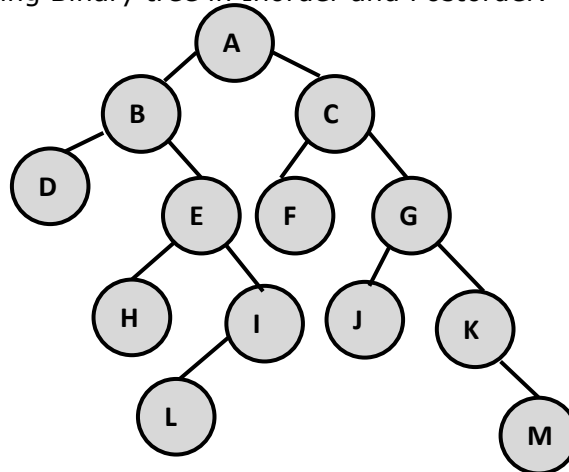
- Write a C function to perform push and pop operations in the stack using a dynamic array. Note: If the stack becomes full, double the stack size. CO2 (08)
 - Apply infix to postfix conversion algorithm to convert the following infix expressions to postfix expressions:
i. $(A*B)/C*D+(E+F)$ ii. $A + (B*C-(D/E)*G)*H$ CO2 (08)
 - Compare and contrast Linear Queue and Circular Queue. CO2 (04)
- Illustrate add and delete operations on a Circular queue with suitable examples. CO2 (06)
 - Use the postfix evaluation algorithm to evaluate the following expression:
i. $6\ 2\ /\ 3 - 4\ 2\ * +$ ii. $2\ 3\ * 5\ 4\ * + 9 -$
Tabulate the input, stack contents and output. CO2 (08)
 - Write a C code to Add and delete an item from the i^{th} Multiple stack. CO2 (06)

UNIT – III

- What is a linked list? How do you create a new node in a linked list using C? CO3 (04)
 - Analyze the advantage of doubly linked list over singly linked list with suitable example. CO3 (08)
 - How do you insert a new node at the beginning of a linked list and explain the process for deleting a specific node from the list. CO3 (08)
- What is a circular linked list? Write C functions to perform the following operations on circular linked list.
i) Inserting node at the front ii) Finding length of a list. CO3 (08)
 - Explain sparse matrix representation using linked list. CO3 (06)
 - Write a C function to invert singly linked list. CO3 (06)

UNIT – IV

7. a) Assuming that the key element is always a leaf node, develop a C function to delete the key element in a Binary Search Tree. Also, Construct the Binary Search Tree for the following data entered as a sequential set. CO4 (10)
"SNAPDEAL", "AMAZON", "MYNTRA", "FLIPKART", "AJIO", "SHOPCLUES", "LIMEROAD".
- b) Traverse the following Binary tree in Inorder and Postorder. CO4 (06)



- c) Write a function for postorder traversal of a binary tree. CO4 (04)
8. a) Given with the following Inorder and Postorder traversals, construct the Binary tree: CO4 (06)
Inorder: D,B,H,E,L,I,A,F,C,J,G,K,M
Postorder: D,H,L,I,E,B,F,J,M,K,G,C,A
- b) Construct the Binary Search Tree for the following data entered as a sequential set. Write the outcome of three tree traversals on BST constructed. CO4 (10)
Data: 125, 24, 80, 15, 145, 90, 25
- c) Write a note on threaded binary trees. CO4 (04)

UNIT- V

9. a) Explain with suitable example how a graph can be represented using adjacency list adjacency matrix. CO5 (06)
 b) Write a C function to perform depth first search on a graph. CO5 (04)
 c) Define height-biased leftist tree. With suitable example explain the creation of height-biased leftist tree. CO5 (10)
10. a) Write a note on weight-biased leftist tree. CO5 (06)
 b) Write a brief note on AVL trees and its usages. CO5 (10)
 c) Write C function to perform BFS on a graph. CO5 (04)
